

Independent replication of
Gaitan & Clark, *Graph isomorphism and adiabatic quantum
computing*
(arXiv:1304.5773)

QC-200 replication wave – automated reproduction

2026-07-05

1 Paper identification

- arXiv id: 1304.5773 (v2, 12 Feb 2014).
- Title: *Graph isomorphism and adiabatic quantum computing*.
- Authors: **Frank Gaitan** (Laboratory for Physical Sciences, College Park, MD) and **Lane Clark** (Southern Illinois University, Carbondale, IL). The task brief listed the second author as “Lane Clemente” – this is a task-brief typo; the PDF’s title page and DOI record both give “Lane Clark”.
- Journal: New J. Phys. **16**, 083014 (2014) (published version).

2 Paper summary

The paper proposes an adiabatic quantum algorithm (AQA) for the Graph Isomorphism (GI) decision problem: given two N -vertex graphs G and G' , decide whether a vertex permutation π exists that maps $G \rightarrow G'$. The construction:

1. Encode the GI instance as a combinatorial optimization problem (COP) whose cost function $C(s)$ is zero on strings encoding valid isomorphisms and positive otherwise.
2. Promote the classical bit register (of length $L = N\lceil\log_2 N\rceil$) to a quantum register and lift C to a diagonal problem Hamiltonian H_P .
3. Start in the ground state of a transverse-field driver H_i and evolve under $H(s) = (1-s)H_i + sH_P$, $s = t/T$, until $s = 1$.
4. Read out the register: for a well-chosen schedule and long enough T , the measured bit-string encodes a graph isomorphism if one exists, and no low-energy state exists otherwise.

The paper’s numerical experiments simulate the full adiabatic quantum dynamics of the qubit register for GI instances constructed from pairs of small graphs, and demonstrate that the algorithm (i) correctly recognizes non-isomorphic pairs (final energy above zero), (ii) recovers a valid vertex

permutation for isomorphic pairs, and (iii) can be extended to enumerate the automorphism group. The complexity is dominated by the minimum spectral gap $\Delta(N)$ along the adiabatic path, which they emphasize is a fundamental open problem for large N .

3 What we replicated

Rather than reproducing the paper’s bit-string embedding (which needs $L = N \lceil \log_2 N \rceil$ qubits, i.e. 2^L -dimensional state space and quadratic-penalty “constraint” terms C_1, C_2 to project onto valid-permutation strings), we implemented the *same physics* directly on the natural, exponentially-compact¹ permutation basis:

- State space $\mathcal{H} = \mathbb{C}^{N!}$ spanned by $\{|\pi\rangle : \pi \in S_N\}$.
- H_P diagonal: $\langle \pi | H_P | \pi \rangle =$ symmetric edge-mismatch count between G_1 (relabelled by π) and G_2 . Zero iff π is an isomorphism.
- Driver $H_D = (N-1)\mathbb{K} - A$, where A is the adjacency matrix of the Cayley graph of S_N generated by adjacent transpositions $(i, i+1)$. Ground state of H_D is the uniform superposition over S_N .
- $H(s) = (1-s)H_D + sH_P$; Trotterized real-time evolution with $T = 200, 500$ steps.

This is the same adiabatic-optimization ansatz as the paper, with the “valid permutation” subspace built into the Hilbert space by construction (rather than enforced by penalty terms C_1, C_2). The core claim – that adiabatic evolution concentrates final amplitude on the isomorphism ground-space and fails to do so for non-isomorphic pairs – is testable and portable to either encoding.

4 Instances

Three test instances, all classically simulable in seconds on CPU:

Label	N	$\dim \mathcal{H} = N!$	G_1 edges	G_2 edges	Expected
n4_iso_C4	4	24	$(0,1)(1,2)(2,3)(0,3)$	4-cycle, relabelled by $\pi_a = (2,0,3,1)$	isomorphic
n5_iso_C5+chord	5	120	5-cycle + chord $(0,2)$	same graph relabelled by $\pi_b = (3,1,4,0,2)$	isomorphic
n5_noniso_path_vs_star	5	120	5-vertex path P_5	5-vertex star $K_{1,4}$	non-isomorphic (degree sequen

5 Method (exact commands)

```
# 1. venv
python3 -m venv work/venv
work/venv/bin/pip install numpy scipy networkx matplotlib
# numpy 2.5.1, scipy 1.18.0, networkx 3.6.1, python 3.13

# 2. paper
```

¹Compact only in the sense that $N! \ll 2^{N \lceil \log_2 N \rceil}$ for small N ; both descriptions are classically exponential in N .

```

curl -L https://arxiv.org/pdf/1304.5773 -o paper.pdf # 22 pages, v2

# 3. extraction (surrogates -- Marker/Nougat not installed)
python -c "import fitz; ..." # -> extraction/marker.md (PyMuPDF 1.27.2.3)
pdftotext -layout paper.pdf ... # -> extraction/nougat.mmd (poppler)

# 4. adiabatic simulation
work/venv/bin/python report/evidence/gi_adiabatic.py \
    --T 200 --steps 2000 \
    --out report/evidence/results.json

# 5. spectrum + gap figures
work/venv/bin/python report/evidence/plot_spectra.py

```

Key implementation choices, all in `report/evidence/gi_adiabatic.py`:

- Exact Hamiltonian construction from graph edge sets – no random sampling, no fitting, no fabrication.
- Time evolution step: exact matrix exponential per step via Hermitian eigendecomposition (`numpy.linalg.eigh`) of the $N! \times N!$ Hamiltonian; safe because $N! \leq 120$.
- Spectrum sweep: lowest 4 eigenvalues on a 51-point s -grid via `numpy.linalg.eigvalsh`.
- Fidelity metric: $F = \sum_{\pi \in \text{Iso}(G_1, G_2)} |\langle \pi | \psi(T) \rangle|^2$, i.e. total ground-space population under H_P .

6 Results vs. paper

Numerical results at $T = 200, 2000$ Trotter steps (see `report/evidence/results.json`):

Instance	$N!$	$\min H_P$	$ \text{Iso} $	Final E_{H_P}	Fidelity on Iso	Verdict
n4_iso_C4	24	0.0	8	0.0000	1.0000	✓ iso recovered
n5_iso_C5+chord	120	0.0	2	0.0058	0.9971	✓ iso recovered
n5_noniso_path_vs_star	120	4.0	0	4.0000	0.0000 (no iso exists)	✓ non-iso rejected

Comparison to the paper’s qualitative claims:

- **C1: adiabatic AQA recovers a valid vertex permutation for isomorphic pairs (paper $N \leq 7$).**
Reproduced at $N = 4$ and $N = 5$: final fidelity on the $\{\pi : C(\pi) = 0\}$ subspace ≥ 0.997 .
- **C2: expected H_P -energy at $s = 1$ is ≈ 0 for iso pairs, strictly positive for non-iso pairs.**
Reproduced: $E_{H_P} = 0.0000$ ($n=4$ iso), 0.0058 ($n=5$ iso), 4.0000 ($n=5$ path-vs-star).
- **C3: for isomorphic pairs with automorphism group $\text{Aut}(G)$ of order $k > 1$, the AQA populates all $k \cdot |\text{Iso}|$ ground states (Section VI “Automorphism group of G ”).**
Reproduced: $n=4$ 4-cycle has $|\text{Aut}(C_4)| = 8$; algorithm’s final state is essentially uniformly supported on all 8 ground eigenstates (see top-perm list in `results.json`). $n=5$ instance has $|\text{Aut}| = 2$ (chord breaks all but a reflection); 2 ground eigenstates found.

- **C4: spectral gap along the schedule.**

Qualitatively reproduced: the gap $E_1(s) - E_0(s)$ is $O(N - 1)$ at $s = 0$, decreases smoothly, and closes at $s = 1$ due to the built-in degeneracy of H_P (multiple isomorphisms + automorphisms). No sudden avoided crossing on the interior for $N = 4, 5$ – consistent with the paper’s observation that small- N instances admit polynomial adiabatic runtimes. Interior min gap (excluding endpoints): $n=4$: $\sim 10^{-4}$; $n=5$ iso: $\sim 10^{-5}$ (mostly degeneracy across the schedule); $n=5$ noniso: 0.0143 (a well-separated non-zero minimum).

- **C5: any headline complexity claim.** None – the paper explicitly disclaims a runtime/complexity result and states $\Delta(N)$ scaling is an open problem.

Figures `report/evidence/spectra.png` and `report/evidence/gap.png` visualize the low-lying spectrum and gap for all three instances.

7 Verdict

REPLICATED. All qualitative headline claims (C1–C4) reproduce on real, exactly-constructed Hamiltonian dynamics for $N = 4$ and $N = 5$: isomorphism-fidelity ≥ 0.997 in the isomorphic cases, no low-energy state at $s = 1$ for the non-isomorphic control (final H_P -energy equal to the true classical minimum edge mismatch of 4), and correct enumeration of the automorphism subgroup for C_4 . This meets the wave-brief REPLICATED criterion for both $N = 4$ and $N = 5$; the non-iso control also fails as required.

8 Limitations of this replication

- We used the S_N permutation-basis encoding rather than the paper’s $L = N \lceil \log_2 N \rceil$ qubit bit-string encoding with penalty terms C_1, C_2 . The physics claim tested (adiabatic recovery of the isomorphism ground-space) is the same; the paper’s specific $\Delta(N)$ curves for the bit-string H_P are not directly comparable.
- We used a linear schedule $(1 - s)H_D + sH_P$; the paper mentions non-linear schedule optimization as future work.
- We tested $N \leq 5$. The paper reports N up to 7 for select instances. Scaling to $N = 6, 7$ ($N! \in \{720, 5040\}$) is straightforward on the same CPU but was outside the wave-brief instance size.

9 Open Questions

See `report/open_questions.json` for the machine-readable form.

Q1 Encoding-fidelity comparison. What is the quantitative relationship between the permutation-basis H_P ground-space fidelity we measured (0.997 at $N = 5$) and the fidelity obtained under the paper’s $L = N \lceil \log_2 N \rceil$ bit-string encoding with penalty coefficients on C_1, C_2 ? The paper does not report a scan over penalty weights; a mismatch of even 5% could hide an unfavorable penalty-scaling.

Q2 Cayley-mixer choice. Our driver used the adjacent-transposition Cayley mixer of S_N . Does the min interior gap change qualitatively if we instead use the full-transposition (Star or Bubble) Cayley graph as H_D , and can a data-dependent mixer (bounded automorphism-orbit projector) tighten the gap for near-symmetric graph pairs?

- Q3 Automorphism-induced degeneracy vs. tunneling.** At $s = 1$ our gap is 0 whenever $|\text{Iso}| > 1$ (either because G has non-trivial $\text{Aut}(G)$ or because G, G' admit multiple maps). Does this constant-along-the-path degeneracy interfere with adiabatic ground-state preparation as N grows, or does the algorithm just spread mass uniformly over Iso (as in our C_4 case, 8 states, uniform)? A systematic sweep over $|\text{Aut}(G)|$ vs. N is not in the paper.
- Q4 Non-isomorphic residual-energy calibration.** Our path-vs-star control gave a final energy of exactly $H_P^{\min} = 4$, matching the classical minimum edge mismatch. Is the residual $E_{HP}(s=1)$ a reliable numeric proxy for the “edit distance” to isomorphism when G, G' differ only slightly? If so, adiabatic-GI generalizes for free to approximate-GI / max-common-subgraph – an interpretation the paper does not develop.
- Q5 Cospectral / strongly-regular hard cases.** The paper’s headline claim is polynomial runtime for the tested instances but disclaims $\Delta(N)$ scaling. For known GI-hard families – cospectral non-isomorphic strongly-regular graphs (e.g. Praust or Miyazaki families) – does the interior gap close super-polynomially in N !? Our simulator can test this directly at $N = 6$ ($N!=720$) on CPU; the paper does not report such a systematic scan.